

Human-Led, AI-Supported 🚀

Practical Tips for Students*

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*The use of AI during coursework must be **expressly permitted** by your module leader. This demonstration is not a substitute for that permission.

Getting in touch 🤝

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Drop in any time/day

Email ahead to confirm that I'm there

*This is what Google Gemini thinks my
office door looks like 🤖👉*



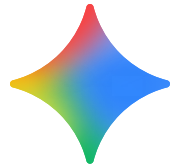
Coming up

1. When can LLMs help?
2. Strengths and Weaknesses of LLMs
3. What is prompting?
4. Building a strong prompt
5. Non-Generative Uses of LLMs 🤖
6. Organising that folder of PDFs
7. ⚡ Power Users: diagrams from `text`
8. ⚡ Power Users: ``code`` blocks
9. ⚡ Power Users: tables as `text`
10. UH Guidance, and Ethical Use of AI

Assuming only free tier access



Copilot



Gemini



Claude



OpenAI

Other LLMs are available: Meta, DeepSeek, Grok (X), Perplexity (multiple models)

Your university Office 365 account provides free tier access to Copilot, with the added protection of an enterprise privacy policy for UH.*

*If you are signed into multiple Microsoft profiles (@outlook.com personal email), this special privacy policy only applies to your @herts.ac.uk profile.

Why and When?

- AI can help you **think, organise, check, and understand**.
- It must not replace **your judgement, your writing, or your academic voice**.
 - *i.e. how you would use **language**; how you would **explain** a theory/concept; **examples** you would draw on; and how you **speak** when interviewed.*
- Using AI is **optional**, and **not needed** for every task.
- Use of AI for coursework is **not permitted unless expressly allowed** by your tutor — read the assignment brief, and locate the relevant AI Declaration cover page
- Use AI to **support** learning, not to produce assessable content.

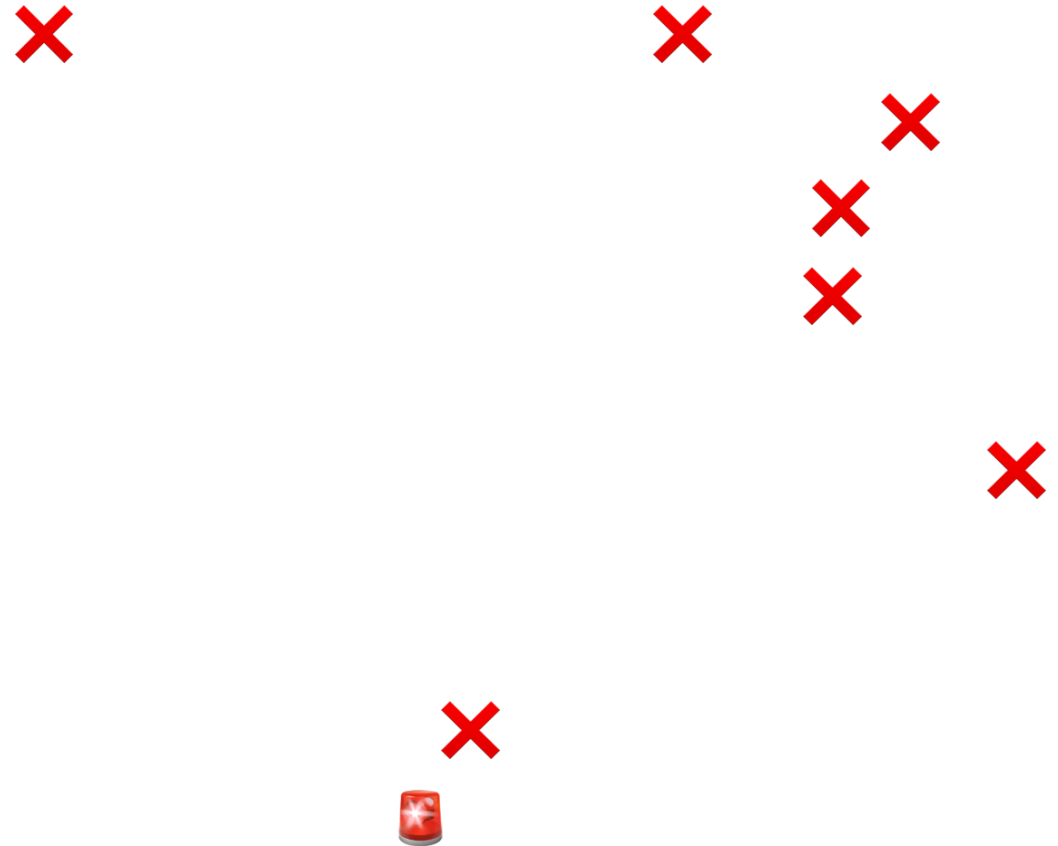
Guiding principle: *Use AI to think with — not to write for you.*

The Good and Bad of LLMs

Strengths

- ✓ Structured programmable tasks
- ✓ Formative and structured feedback
- ✓ Socratic dialogue as learning
- ✓ Synthesis, especially cross-disciplinary
- ✓ Semantic search kills gatekeeping
(missing technical terminology may be
inferred from context and description)
- ✓ Wide choice of models each with **very**
different strengths
- ✓ Use of `markdown` for structured data

Weaknesses



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Weaknesses

- ✗ Will confidently lie you
- ✗ Will hallucinate before admitting failure
- ✗ Will even hallucinate code functions
- ✗ Few models read attachments in full
- ✗ Few models will limit the scope of their responses to your supplied documents
- ✗ Free models have very low model context, meaning your earliest instructions may be silently forgotten
- ✗ **Privacy probably doesn't exist** 🚨

Prompting : Asking AI to do something

Prompting is the act of instructing an LLM to perform a task.

A prompt may be a question or set of instructions, and may include attachments.

The quality of a prompt can significantly affect the response received.

LLMs are **incredibly powerful** when prompted well, but **nearly useless** when prompted poorly.

 The best prompts are **clear, specific, and structured**.

 The worst prompts are **vague, ambiguous, and unstructured**.

What's wrong with these prompts? 🤔

Example Prompt	What's Wrong?
1 "Summarise this."	
2 "Help me do my project."	

Example Prompt	What's Wrong?
1 "Summarise this."	Too vague — no indication of <i>what</i> to summarise, for <i>what audience</i> , or in <i>what format</i> (bullet list, length, key points). This often yields breezy, generic bullet points.
2 "Help me do my project."	Lacks context , clear goal, and structure — <i>project</i> could mean anything; no deliverables, deadlines, or constraints are specified. Models will guess at meaning.

Example Prompt	What's Wrong?
3 "Make a schedule for my week."	
4 "I'm really struggling and I don't know what to do. I'm so stressed about everything."	

Example Prompt	What's Wrong?
3 "Make a schedule for my week."	No context about <i>what should be scheduled, priorities, times, constraints, or objectives</i> . Without context, the model will invent details or produce a generic template.
4 "I'm really struggling and I don't know what to do. I'm so stressed about everything."	Speak to a real person, not AI. Reach out to Student Well-Being (available 24/7 via ask.herts.ac.uk), your programme team (work hours), or HAS-Business for academic skills support for coursework.



Prompting that actually works

CRAFT(e)

Context

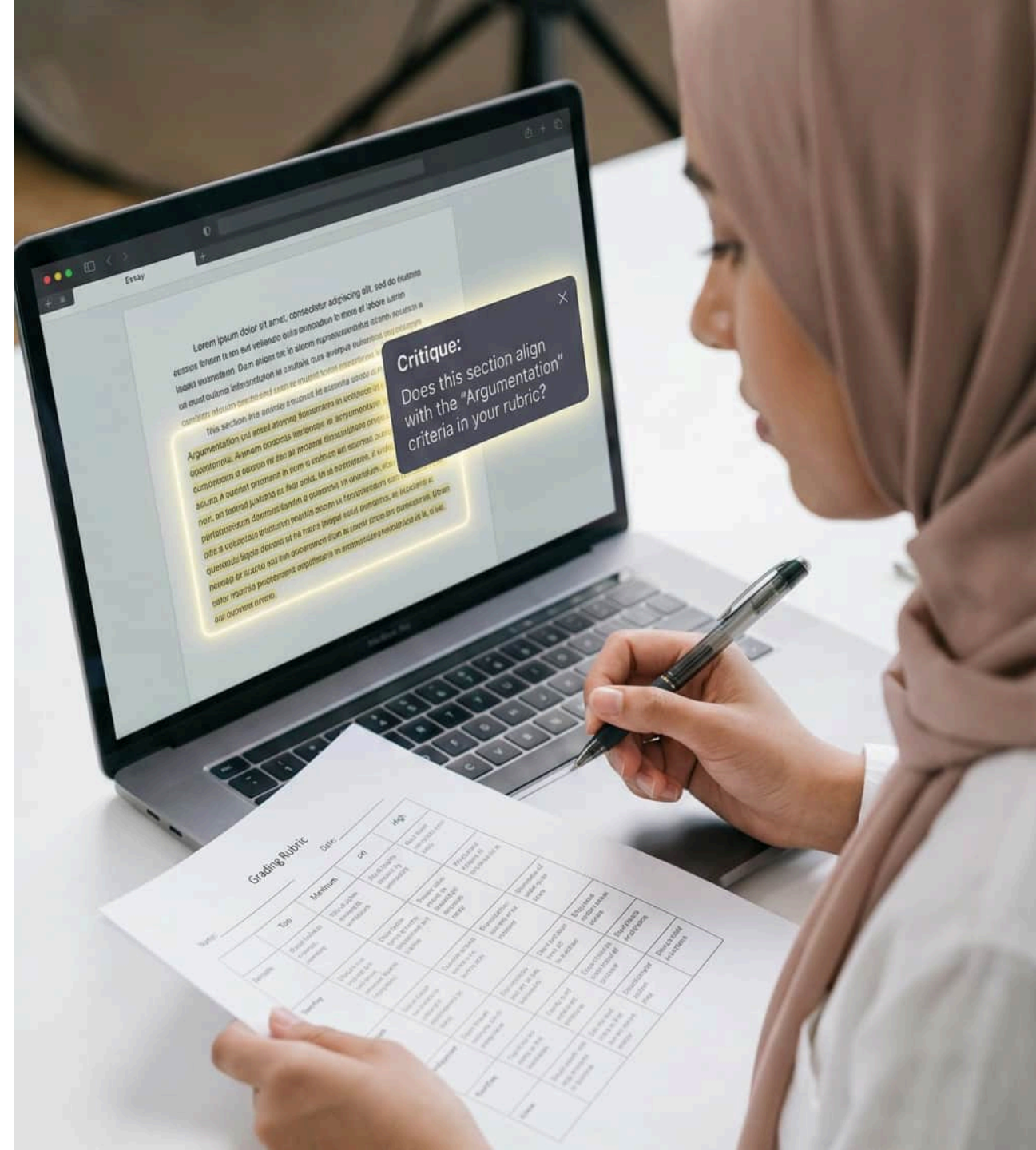
Role

Action

Format

Target audience

(**e**xamples, or templates)



Example Prompt

C ontext, R ole, A ctions

Context:

In the context of studying producer theory in intermediate microeconomics.

Role:

You are a personal tutor helping me to prepare for a test.

Actions:

- 1) Read and index the attached study materials. Recognise diagrams and equations.
- 2) Verify that your index is complete and faithful, and ask me any questions needed to clarify the scope of this task.
- 3) Engage me in a Socratic dialogue about producer theory (30 minutes).
- 4) Create a personalised study plan (2 hours) based on my responses.
- 5) Produce 10 multiple choice questions drawn from the attached study resources only, including relevant diagrams and equations.

Example Prompt

F ormat, T arget audience

Format:

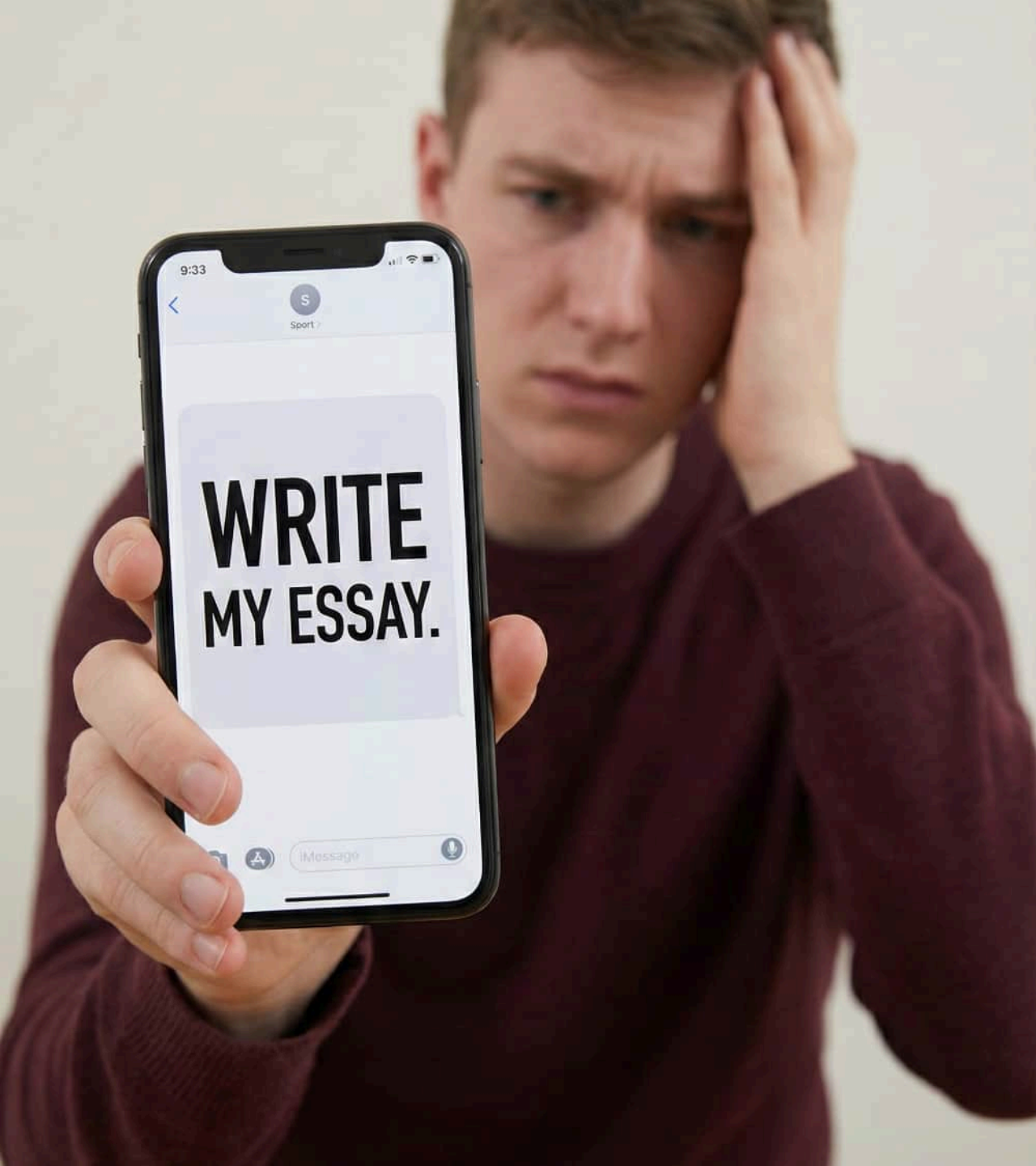
- Respond interactively in this chat
- Use tables for my study plan
- Use multiple choice questions to test me interactively, and provide a download of questions/answers afterwards.
- For each MCQ provide feedback on both the correct and incorrect answers, including page references/links to my uploaded study materials.

Target:

- A second year undergraduate student studying intermediate economics on a bachelor of economics programme at a British university.

Confirmed by Google, in their new prompting guide

- **Break it up.** If you want Gemini to perform several related tasks, break them into separate prompts.
- **Give constraints.** To generate specific results, include details in your prompt such as character count limits or the number of options you'd like to generate.
- **Assign a role.** To encourage creativity, assign a role. You can do this by starting your prompt with language like: "You are the head of a creative department for a leading advertising agency ..."
- **Ask for feedback.** In your conversation with The Gemini app, tell it that you're giving it a project, include all the relevant details, and then describe the output you want. Continue the conversation with questions like, "What questions do you have for me that would help you provide the best output?"
- **Consider tone.** Tailor your prompts to suit your intended audience. Ask for outputs to have a specific tone, such as formal, informal, technical, creative, or casual.
- **Say it another way.** Fine-tune your prompts if the results don't meet your expectations or if you believe there's room for improvement. An iterative process of review and refinement often yields better results.



**OK,
if generated text
is banned, what
else can I do
with LLMs ?**

Formative Conversations (...prompting *it* to prompt *you*...)

As a study advisor...

- Turn readings into **task lists, study plans, revision schedules**.
- Turn lecture notes into **practice questions, flashcards, checklists**.
- Ask for **Socratic questions**, not summaries.

"Engage me in a Socratic conversation about [topic X] to develop my understanding..."

Formative Conversations (...prompting *it* to prompt *you*...)

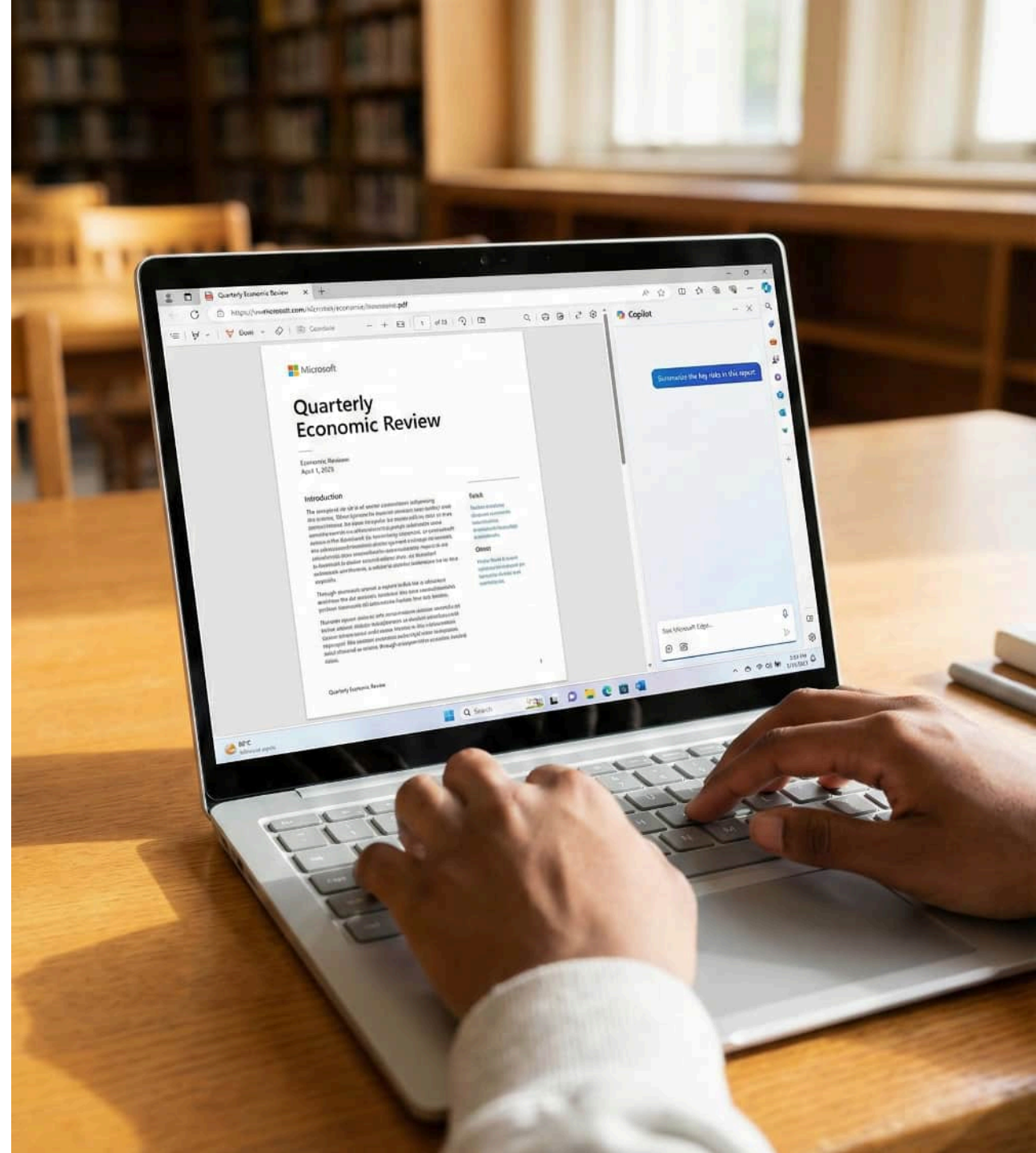
As a writing coach...

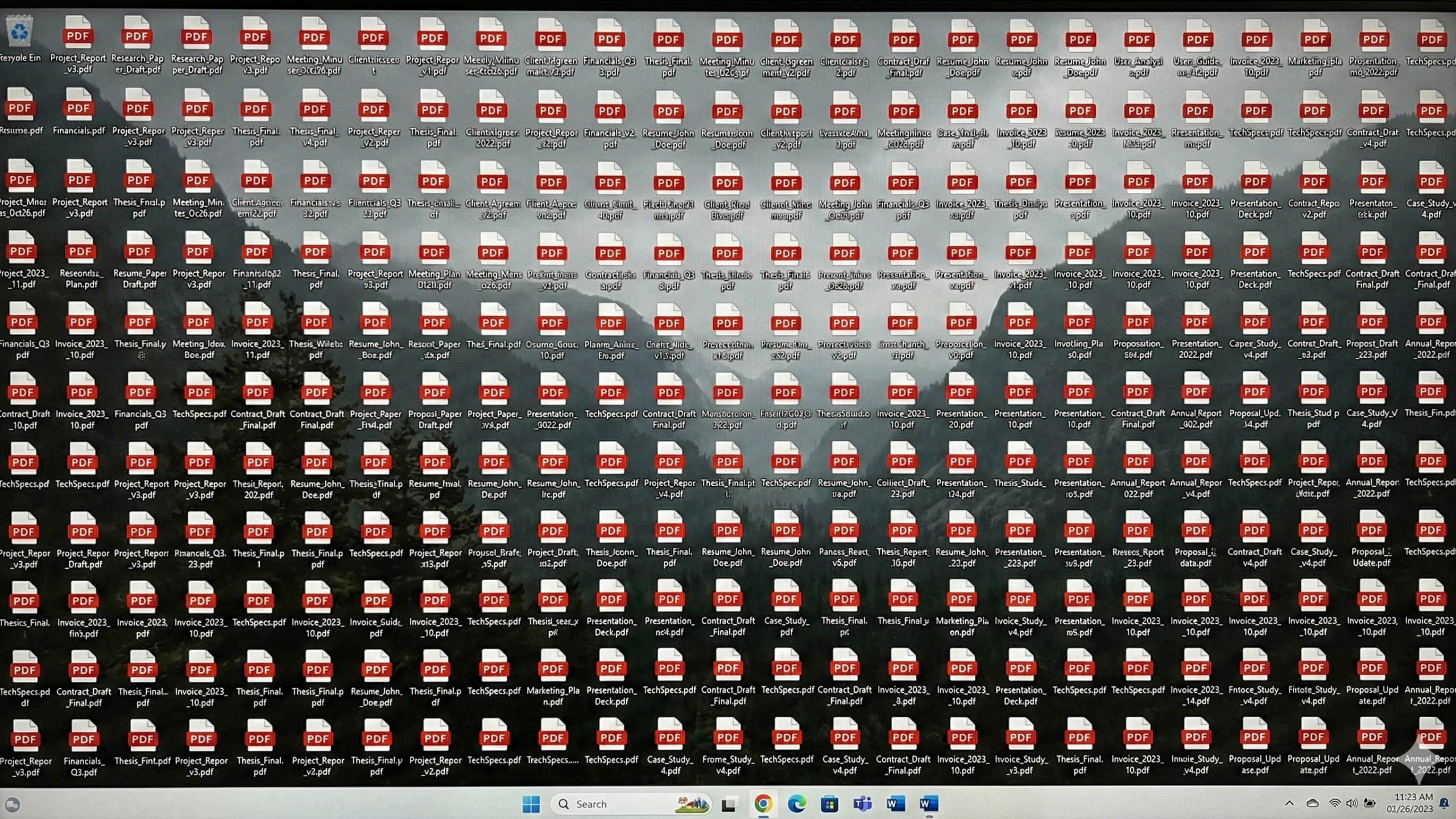
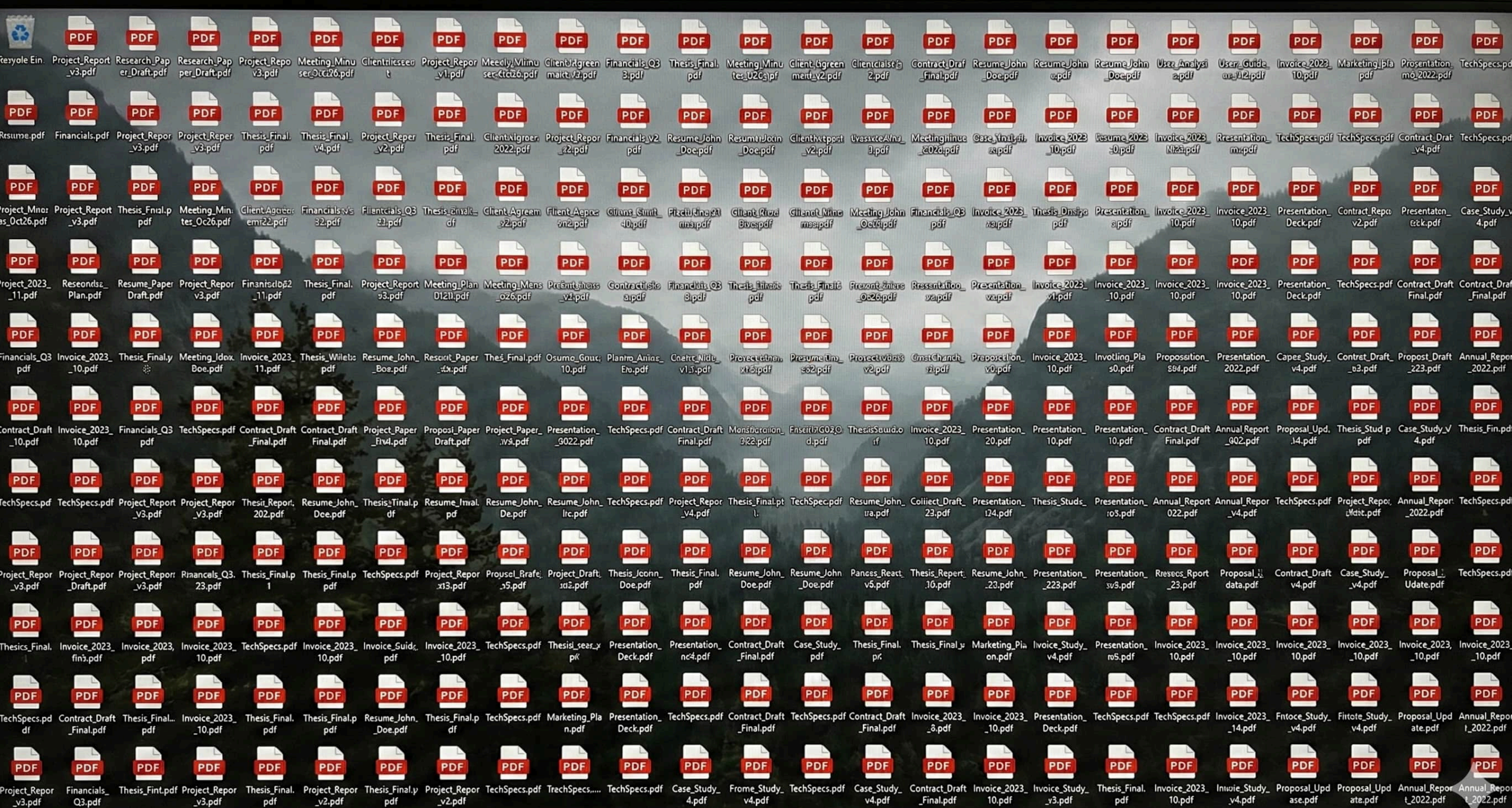
Paste **your own writing**, and ask AI to:

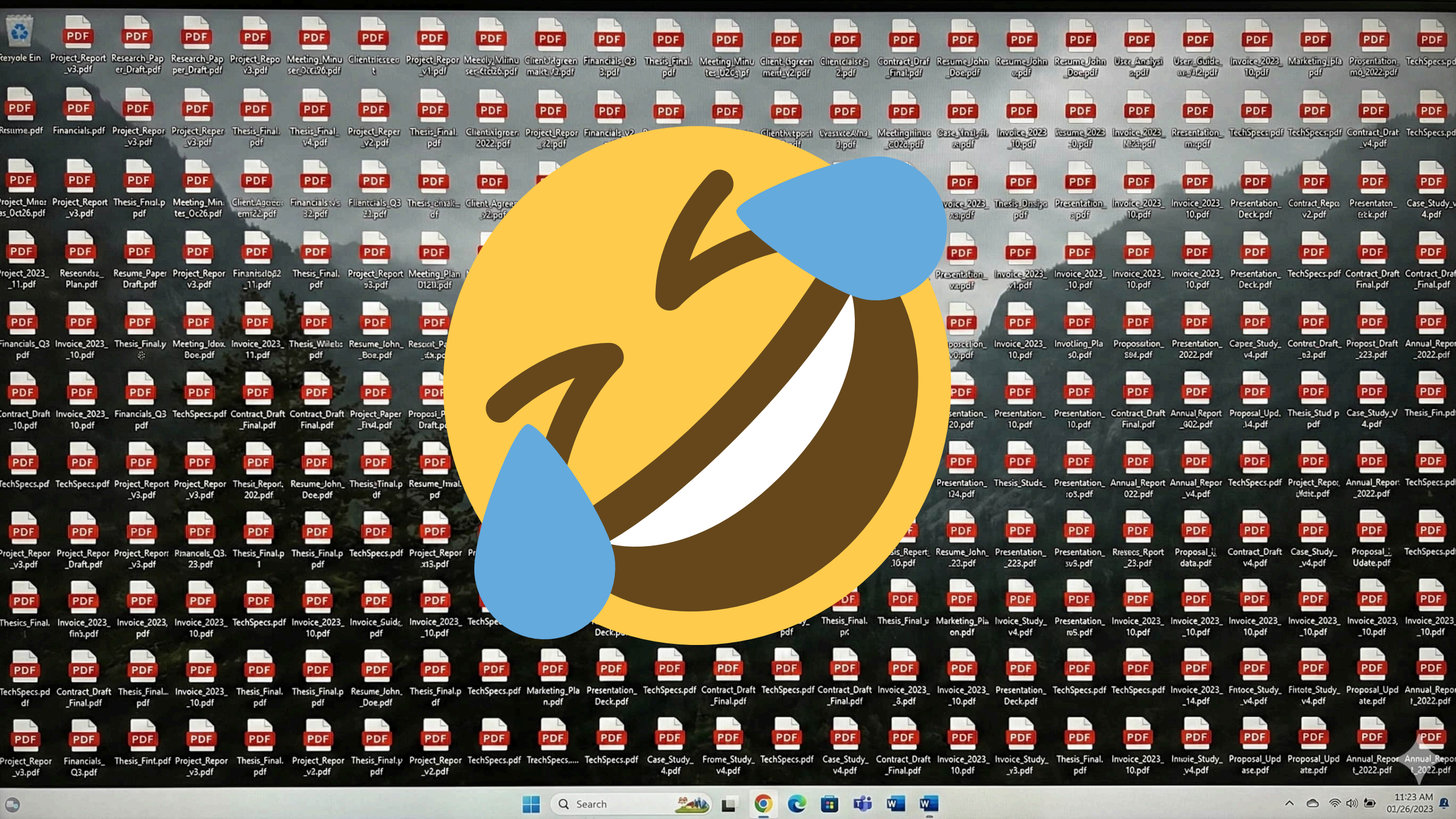
- Identify **YOUR** claims, evidence/premises, conclusions, gaps in your logic or reasoning, your reliance on explicit and implicit assumptions;
- Question the strength of **YOUR** reasoning; prompt you towards further research; prompt you to provide additional counterarguments.

 **Help your future self to avoid academic misconduct by instructing AI to not rewrite your prose; it should support you through questioning, and offering formative advice.**

What to do with
that folder of PDFs ?







Did AI actually read your document? *Probably Not*

Generalist LLMs may **skim** long PDFs
or **miss content** deep within them.

Whereas specialist "RAG" (Retrieval
Augmented Generation) models
confine their responses to just the
information and attachments supplied
by you 💪



Overview of document handling

Feature	NotebookLM (free) 🔥	Claude (free)	ChatGPT (free)	Gemini (free)	Copilot (basic)
Upload PDFs	✓	✓	✓ (limited)	✓	✓
... Multiple	✓	✓	✓ (limited)	✓	✓
Persistence	✓	✗ forgets	✗ forgets	✗ forgets	✗ forgets
Confined Response	✓	Needs prompting	Needs prompting	Needs prompting	Needs prompting
Page refs	✓	Partial	Unreliable	Partial	Partial

Free tier limits change frequently — verify before relying on them.

Google NotebookLM

A RAG tool from Google allowing you to chat with only the information and attachments supplied by you.

- ✓ Responses grounded **only** in your sources — RAG by design, not by prompting
- ✓ Inline citations back to specific passages
- ✓ Persistent **notebooks** — upload once, return later
- ✓ Up to 50 sources per notebook

- i A **folder** of sources you can chat with
- i Ideal for dissertations, literature reviews, large projects with lots of documents
- i Each notebook is a reusable collection — one per module, one per assignment
- ! "Free" with any personal Google account, *when you accept their terms and privacy policy.*

Practical workflow: setting up NotebookLM

- 1 Go to **notebooklm.google.com** and create a new notebook
- 2 Give it a meaningful name — e.g. *"5BUSxxxx Project Literature"*
- 3 Upload your study materials or Google Docs as sources
- 4 The tool indexes them automatically — check the source panel
- 5 Start asking questions — each response cites the sources you supplied
- 6 Come back any time — your notebook and sources persist

Think of each notebook as a module folder you can have a conversation with.

Key takeaways: working with documents

1. **Use the right tool** — NotebookLM for persistent, source-scoped research; Claude or ChatGPT for code creation, execution, interactive canvases; Gemini for images and HTML/JS canvases.
2. **Verify before you trust** — make the AI prove it read the whole document
3. **Constrain explicitly** — in general chatbots, tell the AI to use *only* your sources
4. **Test the boundaries** — ask questions outside the documents to check compliance
5. **Request structured output** — tables and comparisons, not essays
6. **You are the writer** — AI can organise information, but critical judgement and writing is your responsibility. *Use AI as your thinking partner, not your ghostwriter.*

Power Users ⚡

Diagrams from text

LLMs can use plaintext drawing languages like `mermaid` to produce mind maps, timelines, gantt charts, and more.

If your discussion involves obvious levels or categories, or nested information, ask the LLM to produce a mind map using `mermaid`. This will produce the code for a diagram that may be rendered using Obsidian or this website:

<https://mermaid.live>

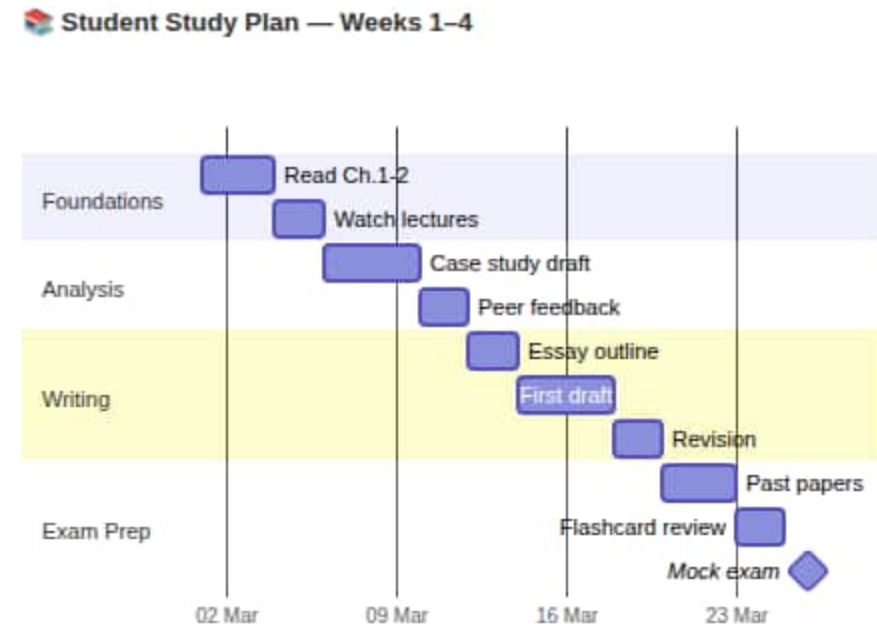
This is an insanely powerful way to visualise complex topics.

⚡ Example: Diagram your study plan

Diagram Code (provided by LLM)

```
```mermaid
gantt
 dateFormat YYYY-MM-DD
 axisFormat %d %b
 tickInterval 1week
 section Foundations
 Read Ch.1-2 :a1, 2025-03-01, 3d
 Watch lectures :a2, after a1, 2d
 section Analysis
 Case study draft :b1, 2025-03-06, 4d
 Peer feedback :b2, after b1, 2d
 section Writing
 Essay outline :c1, 2025-03-12, 2d
 First draft :c2, after c1, 4d
 Revision :c3, after c2, 2d
 section Exam Prep
 Past papers :d1, 2025-03-20, 3d
 Flashcard review :d2, after d1, 2d
 Mock exam :milestone, 2025-03-26, 0d
```
```

Diagram (Rendered)



Rendered using <https://mermaid.live>

Context:

A 4-week university study plan
with tasks organised into sections,
provided below

Role:

Act as a Mermaid diagram expert

Action:

Generate Mermaid gantt chart code
for my study plan.

Format:

- Bare Mermaid code only.
- Use `dateFormat YYYY-MM-DD`
as the only directive

Target:

- Simple and readable
- Minimal syntax
- Suitable for a beginner

...continued

| Section | Task | Start | Duration |
|-------------|------------------|----------------|-----------|
| Foundations | Read Ch.1-2 | 2025-03-01 | 3d |
| Foundations | Watch lectures | after previous | 2d |
| Analysis | Case study draft | 2025-03-06 | 4d |
| Analysis | Peer feedback | after previous | 2d |
| Writing | Essay outline | 2025-03-12 | 2d |
| Writing | First draft | after previous | 4d |
| Writing | Revision | after previous | 2d |
| Exam Prep | Past papers | 2025-03-20 | 3d |
| Exam Prep | Flashcard review | after previous | 2d |
| Exam Prep | Mock exam | 2025-03-26 | milestone |

Code Chunks

Submit code chunks to an LLM by surrounding them in a ````` code fence.

Your keyboard contains a backtick symbol ``` which is not an apostrophe.

```
```r
this is my R code chunk
read.csv(data/mydata.csv)
```
```

Tables as text

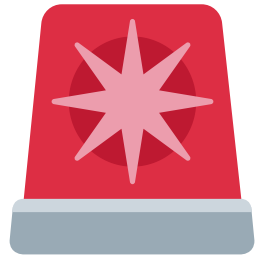
If your LLM can't work with spreadsheet files, ask it to work in `markdown` tables instead.

These are known as "pipe tables" because columns of plain text are separated by `|`.

Use this website to convert between Excel, html and `markdown` tables:

https://www.tablesgenerator.com/markdown_tables

| Header1 | Header2 | Header3 |
|---------|---------|---------|
| Row1 | Row1 | Row1 |
| Row2 | Row2 | Row2 |



Important Advice

1 All LLMs can hallucinate

Older (often free) models hallucinate more; hallucination is a creativity feature, not a bug

Research tools aim for an hallucination tolerance of 0/10

But general LLMs usually tolerate 5/10, which is either **very creative** or **very dangerous** depending on your needs 🤪

2

3

1 All LLMs can hallucinate

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Research tools aim for an hallucination tolerance of 0/10

But general LLMs usually tolerate 5/10, which is either **very creative** or **very dangerous** depending on your needs 🤖

2 Hallucinations in code are easy to spot (code will fail), but hallucinations in academic research are much harder for you to spot

Fake (non-existent) references, fake citations, and fake quotes are serious academic integrity offences

3 ****Do not rely on generative text or generative citations, ever****

"Safer" AI-powered research tools include Scopus AI, Scite.ai and LitMaps.com, but check them too.

UH Resources on the Safe & Ethical Use of AI

<https://go.herts.ac.uk/academic-skills> > Units > AI and Your Studies

<https://go.herts.ac.uk/libraryskillup> > Units > Artificial Intelligence

<https://ask.herts.ac.uk/ai-tools>